Talla Shreyas

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.55
FIITJEE-JUNIOR-COLLEGE	Class XII (Telangana State Board Of Secondary Education,Hyderabad)	2024	9.78%
FIITJEE-WORLD-SCHOOL	Class X (Board of secondary education,Telangana state ,India)	2022	9.80%

PROJECTS

Football Match Outcome Predictor:

- Built and deployed an **end-to-end football match outcome prediction system** using **Python, PyTorch, FastAPI, scikit-learn, Pandas, NumPy, HTML, CSS, and JavaScript**.
- Developed a **feature engineering pipeline** using **historical English Premier League match statistics** and trained a **feedforward neural network** for **multi-class classification**.
- Designed and implemented a **RESTful FastAPI API** integrated with an **interactive web interface** to deliver **real-time predictions**, **probability estimates**, and **team performance comparisons**.
- Architected a **modular application** by separating **data preprocessing**, **model training**, **inference**, **backend services**, and **frontend components**, improving **maintainability** and **scalability**.

Chess Game Engine

- Built a high-performance **Chess Engine** in **C++**, implementing complete chess rules, **legal move generation**, **board state management**, and **UCI (Universal Chess Interface)** support through a modular architecture.
- Implemented advanced game-tree search algorithms including **Negamax**, **Alpha-Beta Pruning**, **Quiescence Search**, and optimized **Move Ordering** to improve search efficiency and engine playing strength.
- Engineered an optimized board representation using **Bitboards**, **Zobrist Hashing**, and **Transposition Tables** for fast move generation, efficient state caching, and accurate position evaluation.
- Developed a configurable evaluation framework utilizing **Piece-Square Tables**, **Material Evaluation**, **Mobility Analysis**, **King Safety**, and **Multithreading** with a custom **Thread Pool**, validating engine correctness through **Perft Testing** and benchmarking.

Choose Your Own Adventure AI

- Built a full-stack AI-powered interactive storytelling platform using **FastAPI**, **React**, **LangChain**, and the **OpenAI API**, implementing a modular **REST API** for dynamic story generation and management.
- Implemented **LLM-based branching narrative generation** with **structured output validation** to create interactive, choice-driven adventures with multiple story paths and endings.
- Developed an **asynchronous backend architecture** with dedicated layers for **story generation**, **business logic**, and **database operations**, improving scalability and maintainability.
- Utilized **PostgreSQL**, **SQLAlchemy**, and **React** for persistent story storage, relational database interactions, and an interactive decision-tree interface for seamless story navigation.

Full-Stack Photo & Video Sharing Application

- Built a **full-stack photo and video sharing application** using **FastAPI**, implementing a **modular REST API** for user management, media storage, and relational database interactions.
- Implemented **JWT (JSON Web Token) authentication and authorization** to secure user endpoints, protect media assets, and manage **user sessions** securely.
- Developed an **asynchronous backend architecture** with dedicated layers for **authentication**, **business logic**, and **database operations**, improving scalability and maintainability.
- Utilized **Pydantic** for **request/response data validation** and modern project tooling with **uv** and **pyproject.toml** to

ensure reproducible development environments.

SKILLS

Languages: C, C++, Python, SQL, HTML,CSS

Frameworks & Libraries: FastAPI, SQLAlchemy, PyTorch, Pydantic, NumPy, Pandas, Matplotlib, Scikit-learn

AI/ML: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, LangChain, Pydantic, RAG, Vector Databases

Tools: Git, GitHub, Docker, Docker Compose, Linux, CMake, VS Code, Jupyter Notebook

Databases: PostgreSQL, SQLite

RELEVANT COURSES

Computer Science: Data Structures, Algorithms, Theory of Computation, Discrete Mathematics, Operating Systems, Database Management Systems

Mathematics: Probability Random Variables, Statistics, Linear Algebra, Differential Equations, Numerical Methods, Complex Variables

POSITIONS OF RESPONSIBILITY

Logistics Coordinator, Media Council

- Coordinated logistics for campus media events, managing resources, schedules, and cross-team communication to ensure seamless event execution.

EXTRACURRICULAR

State Representative (Badminton): Represented Telangana State at the Under-13 National Badminton Championships.

Hostel Competitor (Badminton): Represented the hostel team in the **Milan** inter-hostel sports festival.

Department Athlete: Competed in badminton for the **Diesta** inter-department festival; active participant in recreational cricket and football.